

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

M.Tech. EE Sem – II Examination, 2012-13
EE - 706 Restructuring & Deregulation of power system

Date: 13/05/2013

Time: 10 am to 1 p.m

Maximum Marks: 70

Instructions:

1. Attempt all questions from both sections.
2. Use separate answer book for each sections.
3. Figures to the right indicate full marks.
4. Assume suitable additional data, if necessary

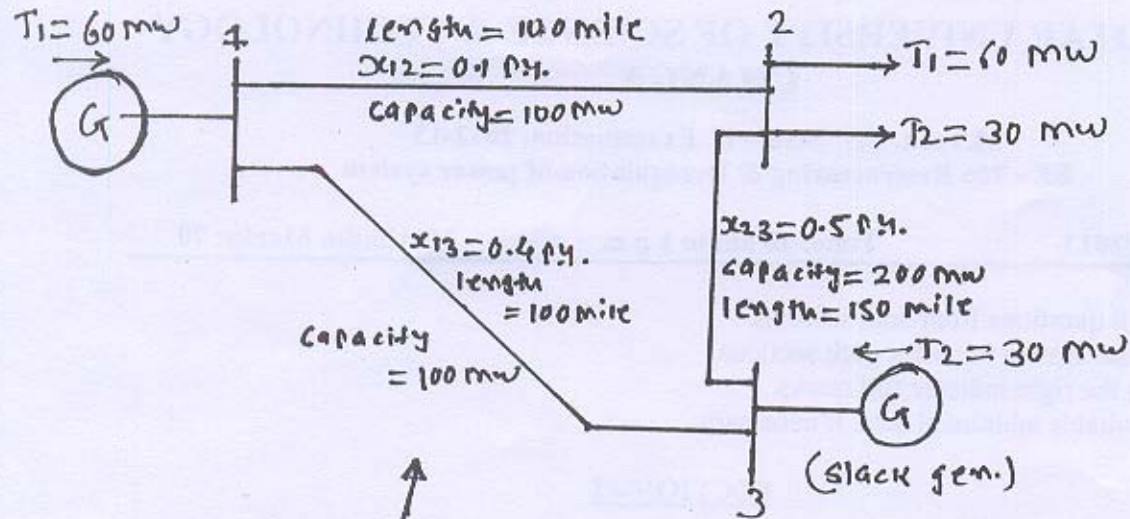
SECTION – I

- Q-1 A. Define Social welfare function. Explain graphically the effect of TCSC on the Consumer, Producer and Merchandize surplus. What is the meaning of negative reactive power spot price? [7]
B. Explain the role of Independent system operator. [5]
- Q-2 A. Write and explain the equation of opportunity cost of reactive power of the synchronous generator [7]
B. Prove that Lerner index $L = S_i / e$ for the oligopoly market, where S_i is the market share of firm i and e is the elasticity of demand [5]
- Q-3 A. Explain the purpose of restructuring of the power system. [6]
B. Define : Cross elasticity of demand, Marginal Revenue and Marginal Cost [5]

SECTION – II

- Q-4 A. For a given 3 bus test system in Fig.1, calculate transmission charges using Rated System Path based method. MW-Mile rate=Rs. 1000/pu MW-Mile hr. [8]
B. Derive the equations of proportional sharing approach to trace active power. [4]
- Q-5 A. Explain L, LL, C and CC models of transmission line for tracing the reactive power [8]
B. Define: Optimal Reactive Power Dispatch. Write different types constraints associated with it. [4]
- Q-6 A. Define TTC. Derive the equations of AC ATC considering ACLODF and VDF factors. [7]
B. Explain Bertrand model for imperfect competition. [4]
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EE-306



(Fig: 1)